

Assessment Brief – Coursework

Key information

Academic Year	2026/2027
Semester	Semester 1
Module Code	CM2604
Module Title	Machine Learning
Assessment Description	Individual Coursework (Two Parts) Part 01: Assignment with Viva Part 02: In Class Test
Deadline (date & time)	Part 01: 15th December 2026 Part 02: At the end of the semester
48-Hour Automatic Extension Available	Yes
Submission	Assessment Dropbox in the Module Study Area in Moodle Attending the viva is compulsory. Failing to attend the viva will result in a 'Fail' grade
Word Limit	2000
Use of Generative Artificial Intelligence (GenAI) See Using Gen AI at RGU	5 Full AI – Part 01 1 No AI – Part 02
Module Coordinator	Sahan Priyanayana

What knowledge and/or skills will I develop by undertaking the assessment?

By the successful completion of this coursework, students will be able to satisfy the requirements that have been set out under learning outcome 1, creating a dataset for ML using data and feature engineering methods applied to a real-world data collection, and learning outcome 3, using ML evaluation methodologies to compare and contrast supervised and unsupervised ML algorithms using an established machine learning framework and learning outcome 4, evaluating ethical and social considerations related to the real world applications and learning outcome 2, critically analyze theoretical concepts and foundations of the machine learning.

On successful completion of the assessment, the student will be able to:

1. To create a dataset for ML using data and feature engineering methods applied to a real-world data collection
2. To critically analyse the theory, including statistical and mathematical underpinning of a range of ML algorithms
3. To use ML evaluation methodologies to compare and contrast supervised and unsupervised ML algorithms using an established machine learning framework.
4. To analyse ethical, social, professional, and legal issues associated with collecting/creating datasets and use of machine learning models in the real world

Please also refer to the Module Descriptor, available from the module Moodle study area.

What is expected of me in this assessment?

Task(s) – content

Part 01 – Individual Assignment

You are tasked with performing a binary classification problem to predict if a customer will "churn" (i.e., cancel their subscription or service) using a telecommunications dataset. The dataset should contain various customer attributes like contract type, monthly charges, and

tenure. Use the ["Telco Customer Churn"](#) dataset, which is publicly available on platforms like Kaggle. The corpus's metadata, class distribution, attributes, attribute statistics, etc., can be found in the provided link.

Task 1

Perform an Exploratory Data Analysis on the dataset and visualize. Document the insights extracted from the analysis in the report

Task 2

Implement the classification task using a KNN, a Logistic regression, and a Random Forest model. Conduct Data preprocessing tasks required for the dataset, problem, and the algorithms. Perform hyperparameter tuning required for algorithms. Include all the information regarding your work and results in the report. Respective libraries, frameworks, tools, etc., must be used for model implementation. The implemented models should be compared based on the optimal evaluation metrics. Experimental results should be showcased for both model experimental settings.

Task 3

What are the strategies you followed in developing these models, considering AI ethics? What are the post-deployment strategies you should follow? Include answers in your report

Part 02: Written In-Class Test

An in-class test will be held to evaluate the understanding of the theoretical concepts.

The date and further details of the test will be provided in due course

Task(s) – format

Part 01 – Individual Assignment

- The implementation must be governed through GIT.
- A report should be prepared, including corpus preparation, solution methodology, evaluation criteria, model evaluation, experimental results, any limitations, and possible further enhancements.
- The project GIT URL should be publicly accessible and should be mentioned in the report. The report should have an appendix that contains all the source code (added as text, not as screenshots).
- If the source code is not added as text in the appendix, it won't be accepted as a valid submission.
- Word limit: 2000 words
- You may use **5: Full AI**. For further details, please refer to the table on page 10.
- There will be an **in-person individual viva** after the submission date.
- Students with **extenuating circumstances** can apply for an **approved, standard five working-day extension** using the Coursework Extension Request. The five working-day extension shall commence from the original submission date. Requests must be submitted to the registry **no later than 24 hours** before the original submission date.

Part 02 – Written In Class Test

This will be an examination paper consisting of **04 questions**

- Duration - **Two (02)** hours
- You must answer all questions.
- This paper carries **60%** weight for CW.

- Students need to come to IIT.
- The registry will inform them of the date, time, and venue.
- This is a **closed-book written examination**, and they should answer the questions in the booklet provided.
- Students may make use of a **non-programmable calculator**.

How will I be graded?

CW Part 01 will be graded according to the rubric provided in the **Assessment Brief**. The mark for CW Part 01 will be **out of 100**.

CW Part 02 will be graded according to the marks allocated for each question in the examination paper. The mark for CW Part 02 will be **out of 100**.

The **Final Module Mark out of 100** will be calculated using the following formula

$$\text{Final Module Mark (\%)} = (\text{CW Part 01 Mark} \times 0.4) + (\text{CW Part 02 Mark} \times 0.6)$$

Finally, the total module mark will be converted to the **Final Module Grade** according to the following criteria.

A **Final Module Mark** $\geq 75\%$

B $75\% > \text{Final Module Mark} \geq 65\%$

C $65\% > \text{Final Module Mark} \geq 45\%$

D $45\% > \text{Final Module Mark} \geq 35\%$

E $35\% > \text{Final Module Mark} \geq 25\%$

F $25\% > \text{Final Module Mark} \geq 0\%$

NS Absent from at least one part of the CW

Feedback Grid

GRADE	EXCELLENT	COMMENDABLE/VERY GOOD	GOOD	SATISFACTORY	BORDERLINE	UNSATISFACTORY
DEFINITION/ CRITERIA (WEIGHTING)	Outstanding Performance	Meritorious Performance	Highly Competent Performance	Competent Performance	Borderline	Fail
Corpus Preparation (25%)	70% - 100%	60% - 69%	50% - 59%	40% - 49%	30% - 39%	0 % - 25%
Implementation (25%)	70% - 100%	60% - 69%	50% - 59%	40% - 49%	30% - 39%	0 % - 25%
Evaluation (25%)	70% - 100%	60% - 69%	50% - 59%	40% - 49%	30% - 39%	0 % - 25%
AI ethics (10%)	70% - 100%	60% - 69%	50% - 59%	40% - 49%	30% - 39%	0 % - 25%
Limitations and Future Enhancements (5%)	70% - 100%	60% - 69%	50% - 59%	40% - 49%	30% - 39%	0 % - 25%

GRADE	EXCELLENT	COMMENDABLE/VERY GOOD	GOOD	SATISFACTORY	BORDERLINE	UNSATISFACTORY
DEFINITION/ CRITERIA (WEIGHTING)	Outstanding Performance	Meritorious Performance	Highly Competent Performance	Competent Performance	Borderline	Fail
Viva Performance (10%)						

Coursework received late, without valid reason, shall not be accepted and shall receive no grade, but shall count as one of the assessment attempts.

What else is important to my assessment?

What is the Assessment Word Limit Statement?

It is important that you adhere to the Word Limit specified above. The Assessment Word Limit Statement can be found in Appendix 3 of the [RGU Assessment Policy](#). It provides detail on the purpose, setting and implementation of wordage limits; lists what is included and excluded from the word count; and the penalty for exceeding the word count.

What's included in the word count?

The table below lists the constituent parts which are included and excluded from the word limit of a Coursework; more detail can be found in the full Assessment Word Limit Statement. Images will not be allowed as a mechanism to circumvent the word count.

Excluded	Included
• Cover or Title Page • Executive Summary (Reports) or Abstract • Contents Page • List of Abbreviations / Acronyms • List of Tables / Figures • Tables – mainly numeric content • Figures • Reference List and/or Bibliography • Appendices • Glossary	• Main Text e.g. Introduction, Literature Review, Methodology, Results, Discussion, Analysis, Conclusions, and Recommendations • Headings and subheadings • In-text citations • Footnotes (relating to in-text footnote numbers) • Quotes and quotations written within “...” • Tables – mainly text content

What are the penalties?

The grade for the submission will be reduced to the next lowest grade if:

- The word count of the submission is above the specified word limit by more than 10%.
- The submission contains an excessive use of text within Tables or Footnotes.

What are the categories of GenAI use?

The table below describes the different categories of GenAI use in assessment.

Category	Description *
1 No AI	GenAI use is not authorised. You must not use GenAI at any point to complete the assessment.
2 Ideas	You may use GenAI to plan, research and generate ideas for the content and structure, or on how to improve the work. - This may be optional or a requirement of the assessment. - No GenAI content is allowed in the final submission.
3 Editing	You may use GenAI to improve the communication and appearance of your work, i.e. to refine, edit and enhance the language or presentation of the original work (e.g. proofreading; image editing). - This may be optional or a requirement of the assessment. - No new ideas can be created. - English is the language of assessment at RGU, therefore submitted work must not include text produced using translation tools (false authorship).
4 Tasks	You can use GenAI to complete certain elements of the assessment task. This may be optional or a requirement of the assessment.
5 Full AI	You can use GenAI throughout the assessment. - This may be optional or a requirement of the assessment. - English is the language of assessment at RGU, therefore submitted work must not include text produced using translation tools (false authorship).

* For all categories, please see the [Academic Integrity in Practice](#) resource which contains guidance on [Document Control](#) for maintaining academic integrity. Keep a copy of all coursework and any material demonstrating how the coursework was produced until after receipt of a confirmed result from the Assessment Board and the period for submission of academic appeals has elapsed.

Citing and referencing use of GenAI

RGU Library has guidance for citing and referencing use of GenAI in the style of [Harvard](#), [Vancouver](#), [APA](#) and [OSCOLA](#).

Acknowledging use of GenAI

If you have to acknowledge GenAI use, you can use this template unless otherwise specified:

Delete as appropriate:

- I have not used GenAI to help me complete my assessment.
- I acknowledge use of [insert name of GenAI tool(s)] from [url of GenAI tool(s)] to help me complete my assessment in the following ways:

Delete as appropriate:

- To help me search for literature related to my assessment.
- To help me generate ideas and/or potential topics for my assessment.
- To help me create a structured outline for my assessment.
- To help me proofread my assessment.
- To help me improve the wording and/or phrasing in my assessment.
- To help me edit the length of my assessment.
- To help in the following ways not covered by the above [student should enter free text].
- I entered the following prompts on [insert date]: [insert prompts] and [insert description of how the generated content was used].

Academic Integrity

For information on academic integrity, please see the Moodle page [Academic Integrity](#) and the resource [Academic Integrity in Practice](#) which contains guidance on [Document Control](#) for maintaining academic integrity. Categories of academic misconduct are defined in [RGU Academic Regulation A3-2 Student Conduct Procedure](#) and include the definitions below.

What is false authorship?

“The University defines this as the practice of submitting work where the student is not the author of that work. The ‘false authorship’ may relate to the student engaging with a third party and/or software tool to complete an assessment, either in part or whole. This may include work produced by, but not attributed to: another student, an essay mill, a family member or friend, a tutoring service or the unauthorised use of Artificial Intelligence (AI) software. It may also include payment, or other favours, though this will not always be the case. It may relate to any form of assessment or conducting research”.

What is plagiarism?

Plagiarism is “the practice of presenting the thoughts, writings or other output of another or others as original, without acknowledgement of their source(s) at the point of their use in the student’s work. All materials including text, data, diagrams or other illustrations used to support a piece of work, whether from a printed publication or from electronic media, should be appropriately identified and referenced and should not normally be copied directly unless as an acknowledged quotation. Text, opinions or ideas translated into the words of the individual student should in all cases acknowledge the original source”.

What is collusion?

Collusion is defined as “two or more students working together, without the prior authorisation of the Course Leader, tutor or supervisor, to produce the same piece of work, and then attempting to present this work as their own”.

What if I’m unable to submit?

- The University operates a [Fit to Sit Policy](#) which means that if you undertake an assessment then you are declaring yourself well enough to do so.
- If you require an extension due to extenuating circumstances, you should complete and submit a Coursework Extension Form. This form is available on the RGU [Student and Applicant Forms](#) page.
- All extension requests must be submitted a minimum of 24 hours in advance of the specified deadline and extensions will be provided for a maximum of 5 working days from the specified deadline.
- You are expected to submit your work by the specified deadline. The University does operate a 48-Hour Automatic Extension which is available to you in the event you need it. The purpose of the automatic extension is to support you with minor illness, unexpected or adverse events that might be impacting your ability to submit your coursework by the submission date. These do not constitute extenuating circumstances.
- There may be limited instances where the 48-Hour automatic extension cannot apply and the Coursework Assessment Brief will state if this is the case in the ‘48-Hour Automatic Extension Available’ box, therefore please check this Assessment Brief carefully. Please also refer to [Regulation A4: Assessment and Recommendations of Assessment Boards](#) and the [Fit to Sit Policy: Extenuating Circumstances](#).
- Further support is available from your Course Leader.

What additional support is available?

- [RGU Study Skills](#) provide advice and guidance on academic writing, study skills, maths and statistics and basic IT.
- [RGU Library guidance on referencing and citing.](#)
- [The Inclusion Centre: Disability & Dyslexia.](#)
- Your Module Coordinator and Course Leader can also provide support.

What are the University rules on assessment?

The University Regulation '[A4: Assessment and Recommendations of Assessment Boards](#)' sets out important information about assessment and how it is conducted across the University.